

IA: MACHINE LEARNING FOUNDATIONS

Object Detection with YOLO

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COMPUTER VISION

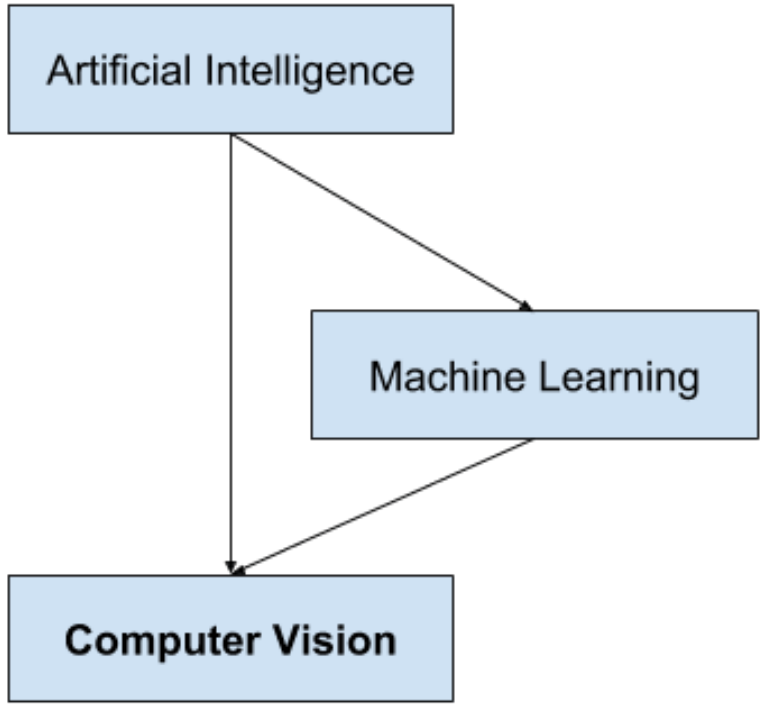
- **Computer vision** is a field of study focused on the problem of helping computers to see.

- Computer vision is distinct from image processing. **Image processing** is the process of creating a new image from an existing image, typically simplifying or enhancing the content in some way.

  - Normalizing properties of the image, such as brightness or color.
  - Cropping the bounds of the image, such as centering an object.
  - Removing digital noise from an image.

- Many popular computer vision applications involve trying to recognize things in photographs:

  - Object Classification
  - Object Detection
  - Object Segmentation



OBJECT DETECTION WITH YOLO

- YOLO is an algorithm that uses neural networks to provide **real-time object detection.**

- Object detection is a phenomenon in computer vision that involves the detection of various objects in digital images or videos. This phenomenon seeks to answer two basic questions:
  1. What is the object? This question seeks to identify the object in a specific image.
  2. Where is it? This question seeks to establish the exact location of the object within the image.

- YOLO is an abbreviation for the term 'You Only Look Once'. This is an algorithm that detects and recognizes various objects in a picture (in real-time). Object detection in YOLO is done as a regression problem and provides the class probabilities of the detected images.

- YOLO algorithm employs convolutional neural networks (CNN) and the prediction in the entire image is done in a single algorithm run.

- YOLO algorithm is important because of the following reasons:
  - **Speed**: This algorithm improves the speed of detection because it can predict objects in real-time.
  - **High accuracy**: YOLO is a predictive technique that provides accurate results with minimal background errors.
  - **Learning capabilities**: The algorithm has excellent learning capabilities that enable it to learn the representations of objects and apply them in object detection.

OBJECT DETECTION WITH YOLO

V1 ARCHITECTURE

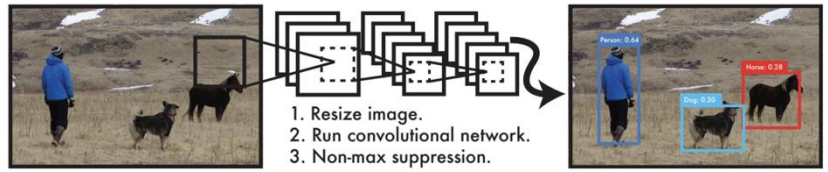
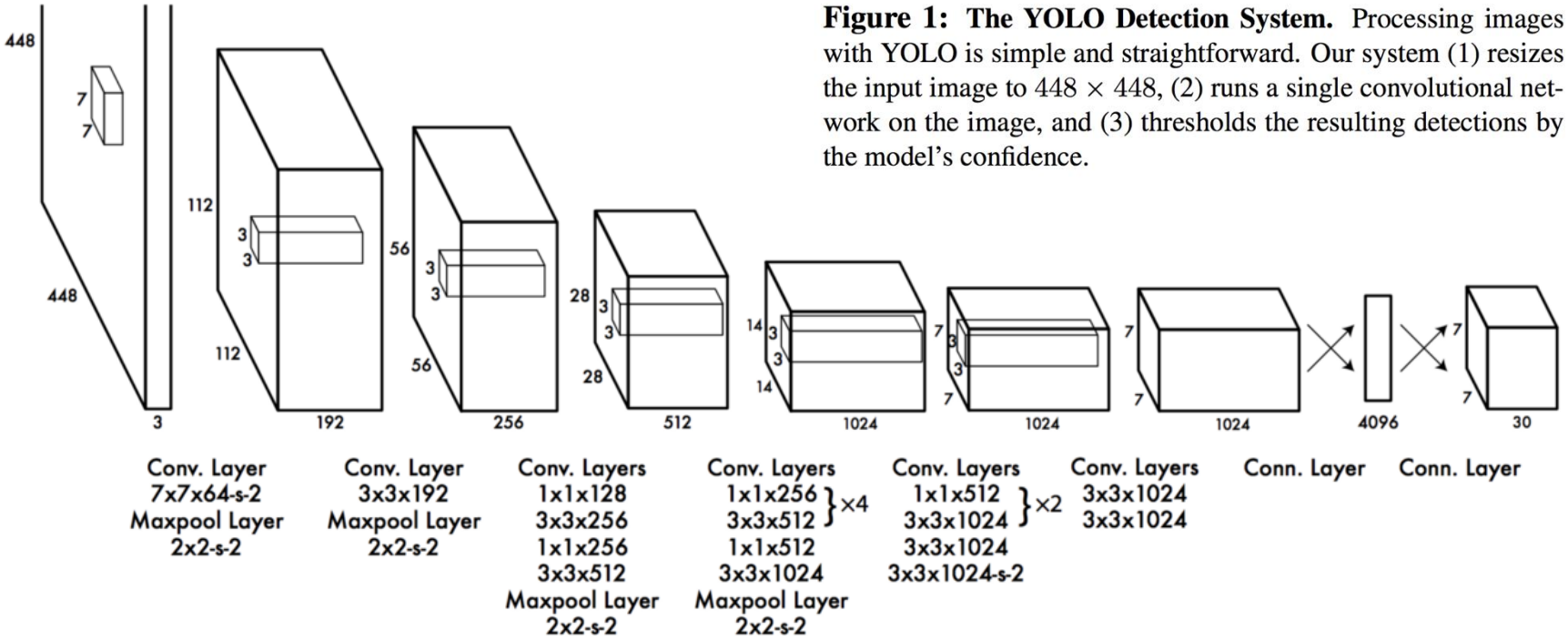


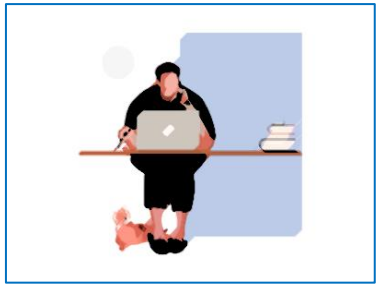
Figure 1: The YOLO Detection System. Processing images with YOLO is simple and straightforward. Our system (1) resizes the input image to 448 × 448, (2) runs a single convolutional network on the image, and (3) thresholds the resulting detections by the model's confidence.

24 convolutional layers followed by 2 fully connected layers



OBJECT DETECTION WITH YOLO

PRACTICE



PROBLEM

Sleepers and Clips detection with YOLO

